Reading a Strategy Performance Report

How to tell from a backtest report whether a strategy is **bad**, **good** or **great**, with the thresholds this project uses (NQ / ES futures, MultiCharts reports, Apex accounts and a future personal account).

Before anything else: every number in this guide is **after costs** and **per 1 full contract** (1 NQ = 10 MNQ). Our MultiCharts backtests usually run with commission and slippage at 0 (Python adds them), so subtract the round-trip cost per trade yourself: **NQ $17.25** ($7.25 commission + 1 tick on entry and exit), **ES $30**, CL / GC $34.50. A strategy that only works before costs is not a strategy.

1. The 30-second triage

If any of these fails, stop: the rest of the report doesn't matter yet.

- **Costs:** is the average trade at least **3x the round-trip cost** (NQ: about $50+ per trade)? Below 2x, a little extra slippage live wipes it out.
- **Enough trades:** at least **100**, better 300+. Twenty great trades prove nothing.
- **Profit factor after costs** above about **1.2**.
- **Years:** positive in most calendar years, not one great year carrying the rest (Periodical Returns, Annual).
- **Settings sheet:** the inputs, bar size, session and time zone are the ones you meant. Old inputs survive a code paste in MultiCharts, and that has broken backtests here before.

2. The main numbers: bad / good / great

Where to find each one in the MultiCharts report is in the second column. t-stat, Sharpe, Calmar and the year count are quick to compute from the List of Trades sheet (or ask Claude: *analysis/metrics.py*).

Metric	Where (MultiCharts)	Bad	Good	Great	Why it matters
Total trades	Performance Summary	< 100	300+	1,000+	More trades = less luck. The t-stat grows with the square root of the count.
Average trade (after costs)	Avg Trade, minus cost	< 2x cost (NQ < $35)	$75-200 per NQ	$250+ per NQ	Room for real-world slippage and fills.
Profit factor (after costs)	Profit Factor	< 1.15	1.25-1.4	1.5+ (2+ = exceptional)	Gross win / gross loss. Above 3 on hundreds of trades: look for a bug.
t-stat = avg / st.dev x √trades	compute from List of Trades	< 2	2.5-3.5	3.6+ (our multiple-testing bar)	How sure the average isn't zero. The bar rises with every idea tested (TRIALS.csv).
Years positive	Periodical Returns: Annual	6/9 or fewer	7/9	9/9	An edge should show up in most years, not one.
Recent period (2023+) vs earlier	split the List of Trades	PF falls below ~1.1	PF within ~0.1	as good or better	The best stand-in for 'will it keep working'.
Sharpe (daily, x √252)	compute; MC's is monthly	< 0.5	0.8-1.5	1.5+	Return per unit of day-to-day swing. Above 3: suspicious.
Sortino	Performance Summary	< 0.8	1.2-2	2+	Like Sharpe but only counts down days; ~1.5-2x Sharpe is normal.
Calmar = yearly profit / max drawdown	Return on Max Strategy DD / years	< 0.5	1-2	2+	How long a worst drawdown takes to earn back.

Metric	Where (MultiCharts)	Bad	Good	Great	Why it matters
Max drawdown at the size you trade	Max Strategy Drawdown	> your limit	< ~60% of it	< ~40% of it	Apex: $2,000 EOD trailing. Size so the backtest DD (and a Monte Carlo 95% DD) fits.
Longs vs shorts	Long / Short columns	one side loses	both positive	both PF 1.3+	On NQ, longs ride the bull market; a long-only edge must beat random long days.
Top 5% of trades / net profit	List of Trades, sort by profit	> 100%	< 70%	< 50%	If the best few trades are the whole profit, a quiet year loses money.

Win rate on its own says nothing. 35% winners with winners 3x the losers is fine; 80% winners with losers 6x the winners is a time bomb. Check **expectancy = win% x avg win - loss% x avg loss** (= the average trade). For Apex, a **high win rate** helps: long losing streaks hit the $2,000 trailing drawdown (look at Max Consecutive Losers x Avg Losing Trade at your size).

3. Red flags (signs of a curve fit or a bug)

- **Many inputs, odd values** (a 37-minute window, a 1.83 multiplier) and results that change a lot when one input moves one step. Good strategies sit on a **plateau**: neighbouring settings work almost as well.
- **Only one period tested**, or the parameters picked on the same period the report shows. Ask for a 2018-22 / 2023+ split and a walk-forward.
- **Costs off** (Commission 0 / Slippage 0 in the Settings sheet) and an average trade of a few ticks.
- **Fills too good:** limit orders filled just by touching the price (we require a trade-through), stop and target in the same bar without Bar Magnifier, or entries at a bar's close using that same close (look-ahead).
- **Session traps:** SetExitOnClose on a 24-hour chart exits at midnight, not 15:59; holidays and half-days trading into a 12:00 CT close; continuous-contract roll jumps (03:30 / 17:00 CT) inside a trade.
- **MultiCharts' 'Adjusted Net Profit' or 'Select Net Profit' far below Net Profit**, or negative: the profit comes from outliers.
- **A Sharpe above 3, PF above 3 or a straight-line equity curve** on hundreds of trades: almost always a bug.
- **Max drawdown is recent and the deepest ever**: the edge may be fading (watch it, don't panic).
- **The report shows great numbers on a source's word** ('proven', 'battle-tested', 'validated'): only our own test on our data counts.

4. What a report can't tell you (the extra checks we always run)

Check	Question it answers	Pass
Random-entry placebo	Does the signal beat the same trades on random days / with a random side?	p <= 0.05
Out-of-sample / walk-forward	Does it work on data it wasn't designed on?	2023+ positive, similar PF
Multiple testing (trials.py)	Is it just the best of many tries?	t >= the bar (3.6 now) or deflated Sharpe >= 0.95
Monte Carlo (montecarlo.py)	How bad can the drawdown get in another ordering of the same trades?	95% DD fits your limit; P(hit $2,000 trail) < ~5% at plan size
Independent checker	Does a fresh rebuild from the rules alone give the same trades, with no leaks?	PASS / PASS WITH CONCERNS
Plan fit (bootstrapped bench)	Does it add to (or replace part of) what we already trade?	faster / more payouts without blown +1 pt
Live tracking (drift alarm)	Is the live average at least ~75% of the backtest?	no ALARM after ~40 trades

5. Putting it together: the tiers

Tier	Rough numbers (after costs)
S	about $20,000+ a year per NQ, t 4.5+, PF 2+, 8/9 years positive, liquid enough to size up
A	t 2.5+, PF 1.4+ (1.9+ if it trades rarely), 7+ of 9 years positive, MultiCharts-confirmed
B	real, profitable edge: t 2-3.5 or PF 1.25-1.4, most years positive
C	positive but small or inconsistent: t 1.5-2.5, weak recent years
D	the effect exists but makes about nothing after costs / stops / rules
F	no edge, loses, flawed study, or can't be traded

6. Worked example: Relative Strength NQ / ES (2026-10-20)

Check	Value	Verdict
Trades	1,161 (2018-26)	great
Average trade after costs	$306 per NQ (~18x the $17.25 cost)	great
Profit factor	1.36 (longs 1.46, shorts 1.26)	good
t-stat	4.0 (bar 3.6)	great
Years positive	9/9	great
2018-22 vs 2023+	$271 vs $344 a trade	great
Sharpe / Sortino / Calmar	1.33 / 2.23 / 1.39	good
Max drawdown	$28.5k per NQ = $2.9k per MNQ	too big for Apex at 1 MNQ alone
Top 5% of trades	111% of the profit	red flag
Placebo / checker	random side p 0.002 / PASS WITH CONCERNS	pass
Result	B (PF just under A's 1.4); concerns: tail-dependence, 9:00-only, weak shorts	

7. Checklist

- Settings sheet: inputs, bar size, session, time zone, Bar Magnifier, commission / slippage as intended.
- Subtract costs: average trade >= 3x the round trip.
- Trades >= 100 (300+ preferred).
- Profit factor after costs >= 1.25 (great 1.5+).
- t-stat >= 2.5 (great: above the multiple-testing bar, 3.6 now).
- Positive in >= 7 of 9 years; 2023+ about as good as 2018-22.
- Sharpe >= 0.8 (daily), Calmar >= 1.
- Max drawdown at your size fits your limit with room (Apex: well under $2,000).
- Longs and shorts each checked; a long-only NQ edge beats random long days.
- Top 5% of trades < ~70% of the profit; Adjusted Net Profit still positive.
- No red flags from section 3.
- Then the extra checks (section 4): placebo, holdout, trials bar, Monte Carlo, checker, plan fit, live tracking.

Thresholds are rules of thumb from this project's testing standard (CLAUDE.md, DEVELOPMENT_PROCESS.md) and tier list (TIER_LIST.md), not universal laws. Backtests are hypothetical; live results usually come in below them (published anomalies lose roughly a quarter to a half of their edge out of sample).